# WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

Twelfth Weekly Report

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# Classification Results - Gesture

**Problem Definition:**

- Gesture recognition is modeled as a **6-class classification** problem.

**Performance Evaluation:**

| Model | Test Mode | Test Accuracy |
|---|---|---|
| Our from scratch | random | 98% |
| Our pretrained finetuned | random | 99% |
| Our pretrained LinearProbing | random | 96% |
| Our from scratch | inter-session (subject S1) | $18\% \pm 3\%$ |
| **Our pretrained finetuned** | **inter-session (subject S1)** | **$70.5\% \pm 7\%$** |
| Our pretrained LinearProbing | inter-session (subject S1) | $60\% \pm 11\%$ |
| CNN | random | 98% |
| CNN | inter-session (subject S1) | $56.5\% \pm 15\%$ |

# Classification Results - Position

**Problem Definition:**

- Position recognition is modeled as a **3-class classification** problem.

**Performance Evaluation:**

| Model | Test Mode | Test Accuracy |
|---|---|---|
| Our from scratch | random | 99% |
| Our pretrained | random | 99% |
| Our from scratch | inter-session (subject S1) | 42% $\pm$ 19% |
| **Our pretrained** | **inter-session (subject S1)** | **81% $\pm$ 11%** |
| CNN | random | 97% |
| CNN | inter-session (subject S1) | 76% $\pm$ 12% |

# Finetuning: Some Trends

- Validation has been performed using an **extra session hold out** (unseen during training).

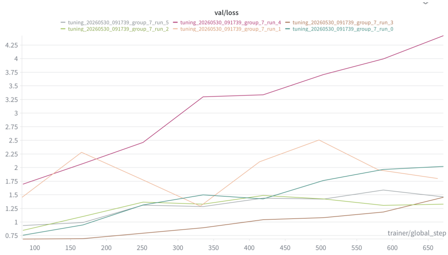


Figure: Loss Trend

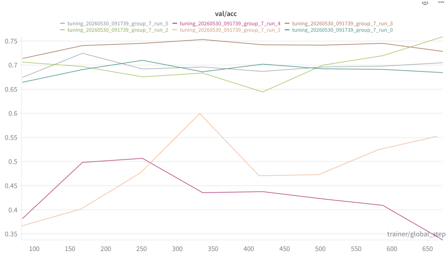


Figure: Accuracy Trend

With a 70-30 validation split across 5 sessions and one hold-out session for testing, this architecture achieves **70.5% accuracy**.

# Linear Probing: Some Trends

- Validation has been performed using an **extra session hold out** (unseen during training).

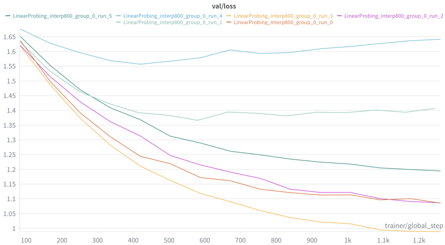


Figure: Loss Trend

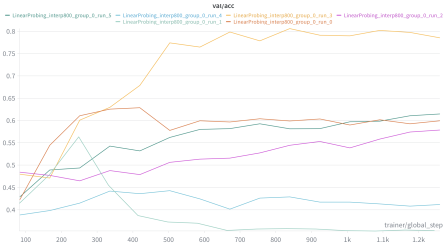


Figure: Accuracy Trend

With a 70-30 validation split across 5 sessions and one hold-out session for testing, this architecture achieves **60% accuracy**.

- Validation has been performed using an **extra session hold out** (unseen during training).


Figure: Loss Trend

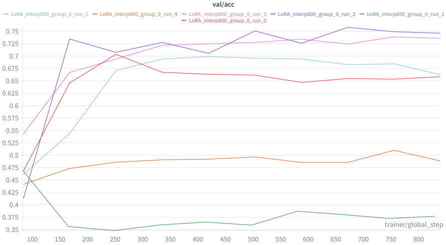


Figure: Accuracy Trend

With a 70-30 validation split across 5 sessions and one hold-out session for testing, this architecture achieves **60% accuracy**.

# Finetuning: Embeddings and Confusion Matrix

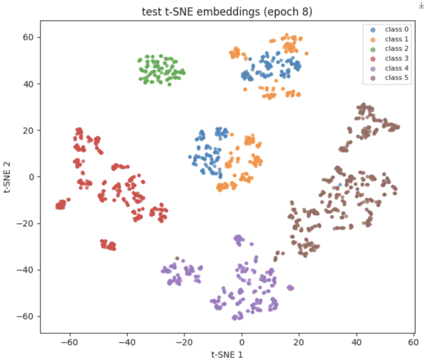


Figure: Latent Space Embeddings

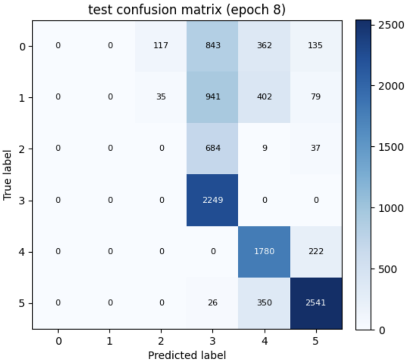


Figure: Confusion Matrix

## Note on Performance Bound

This represents the worst run among the 6 inter-session evaluations, where the model reaches its lower bound of **60% test accuracy**.

# Finetuning: Embeddings and Confusion Matrix

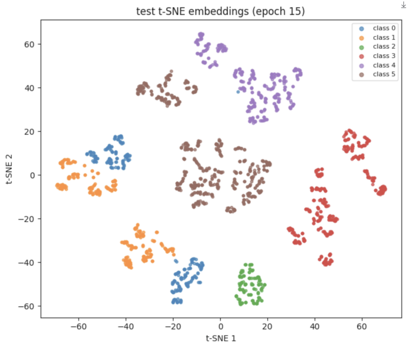


Figure: Latent Space Embeddings

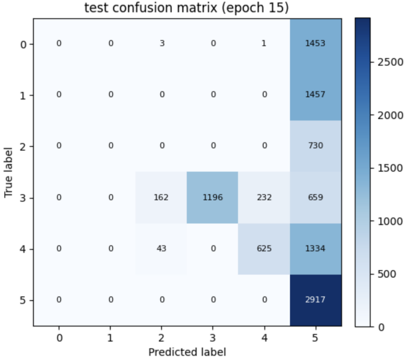


Figure: Confusion Matrix

## Note on Performance Bound

This represents the worst run among the 6 inter-session evaluations, where the model reaches its lower bound of **42% test accuracy**.

# HWT Results

| **Run / Configuration** | **n sessions** | **MAE inter-session** | **Std** | |
|---|---|---|---|---|
| Our pretrained finetuned | 9 | 8.032 | 0.816 | |
| Our fromScratch x4 smaller | 9 | 8.524 | 0.926 | |
| Our fromScratch | 9 | 9.017 | 0.914 | |
| Pau pretrained | – | 7.680 | – | – |
| Pau from-scratch | – | 7.650 | – | – |